25901112 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBTG200 - BIOSAFETY AND BIOETHICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Which BSL requires a full-body, air-supplied positive pressure suit? [K] / [CO3]
2. List any two types of risks faced by an organization. [K] / [CO4]
3. What is informed consent? [U] / [CO5]
4. What is "Analgesia"? [K] / [CO5]
5. Indicate the process used to prevent disease spread. [U] / [CO1]
6. Mention one function of ethics committees in biological research. [U] / [CO1]
7. Name the organization responsible for issuing the Declaration of Helsinki. [U] / [CO5]
8. What is the term for the voluntary agreement given by a minor to participate in research? [U] / [CO5]
9. Identify the equipment used for sterile airflow in Laboratory. [U] / [CO2]

10. Identify any two types of hand protection. [U] / [CO3]

11. Which ethical term refers to data protection? [U] / [CO5]

12. Who has the power to shut down an unethical study? [U] / [CO5]

[2x10 = 20]

Part B

Short Essay

Answer any **4** questions

Each question carries **5** marks

13. Apply ethical principles to propose a strategy that minimizes the use of animals in research. [A] / [CO5]

14. Describe the difference between institutional review board and ethic committees. [U] / [CO5]

15. Discuss five major types of hazards associated with laboratory or industrial workplaces. [An] / [CO4]

16. Explain the purpose of the Cartagena Protocol on Biosafety. [U] / [CO3]

17. Explain the role of communication systems during biosafety incidents. [K] / [CO1]

18. Expalin different physical containment measures. [U] / [CO4]

[5x4 = 20]

Part C

Essay type questions

Answer any **2** questions

Each question carries **15** marks

19. Explain how experimental plant research can contribute to sustainable agriculture. [E] / [CO5]

20. Identify the regulatory guidelines governing rDNA research in India and state their significance. [K] / [CO2]

21. Describe in detail the SOP for aseptic techniques and microbial culture handling in the laboratory. [U] / [CO2]

22. Explain the major components of a biosafety cabinet and their functions. [K] / [CO3]

[15x2 = 30]